3 August 2026

Dear Editor,

We wish to submit "Adult neurogenesis promotes pattern separation and sparsity beyond the dentate gyrus in a spiking DG–CA3 model" for consideration as a Research Article.

Existing models of adult neurogenesis and pattern separation stop at the dentate gyrus. Using a spiking dentate gyrus–CA3 network built from Hippocampome.org and extended with immature granule cells, we show that neurogenesis makes activity sparser and improves pattern separation in CA3 as well, reproducing *in vivo* observations (McHugh et al. 2022) from the known circuitry alone. This extends work the journal has published (Kim and Lim 2024a, b; Yang et al. 2025).

The manuscript is original, is not under consideration elsewhere, and all authors have approved it. It derives from the first author's master's thesis at the Federal University of Paraíba. Code and data: `https://github.com/MarlonVCendron/neurogenesis`. No funding was received, the authors declare no competing interests, and the study is entirely computational.

Sincerely,

Marlon Valmórbida Cendron (corresponding author)
Federal University of Paraíba, João Pessoa, Brazil
marlonvcendron@gmail.com